Chunlei Li (李春蕾)

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Summary

I'm a PhD candidate at the VR Lab of Beihang University from 2021, supervised by Prof. Qinping Zhao (赵沁平), Prof. Yang Gao (高阳), and Prof. Shuai Li (李帅). My research focuses on physical simulation in computer graphics. I was a visiting student at University College London, supervised by Prof. He Wang (王鹤). I expect to graduate in December 2026. Interest: Numerical Methods, Physical Simulation, Physical Based Animation, Reinforcement Learning, Computer Graphics, AI4S, PDE. **Personal homepage: https://chunleili.github.io/**.

Education

University College London, Visiting PhD in Computer Science — Sept 2025 – Sept 2026

Beihang University (BUAA), PhD in Computer Science — Sept 2021 – Dec 2026

Beihang University (BUAA), MSc in Power Engineering and Engineering Thermophysics — Sept 2018 – June 2021

University of Michigan, Dearborn, Exchange Student in Energy and Power Engineering — Sept 2016 – May 2017

North China Electric Power University (Beijing), BSc in Energy and Power Engineering — Sept 2014 – July 2018

Publications

RLMuscle: Muscle-Driven Character Animation via Reinforcement Learning — Oct 2026
Chunlei Li, Siyuan Yu, Yang Gao, Shuai Li, Peng Yu, Aimin Hao, He Wang
https://chunleili.github.io/project-page-rlmuscle/ (arXiv preprint)

Motivation: in most 3D animation pipelines artists hand-tune per-muscle activations, which is time-consuming and lacks physical realism. Novelty: an RL controller trained on a 1D Hill-type surrogate automatically discovers per-muscle activations, which then drive a GPU multigrid-XPBD volumetric simulator with Hill-type fiber constraints, animating a 194-muscle full-body model at 18 ms per frame.

MGPBD: A Multigrid Accelerated Global XPBD Solver — Aug 2025
Chunlei Li, Peng Yu, Tiantian Liu, Siyuan Yu, Yuting Xiao, Shuai Li, Aimin Hao, Yang Gao, Qinping Zhao
https://doi.org/10.1145/3721238.3730720 (SIGGRAPH (CCF-A) | https://chunleili.github.io/project-page-mgpbd/)

Motivation: XPBD's Gauss-Seidel solver stalls on low-frequency errors, causing instability in high-resolution, high-stiffness simulations. Novelty: a global UA-AMG preconditioned CG solver with lazy prolongator reuse and lightweight near-kernel construction, greatly improving convergence and stability at low overhead.

Parallel Constraint Graph Partitioning and Coloring for Realtime Soft-Body Cutting — Apr 2025
Peng Yu, Ruiqi Wang, **Chunlei Li**, Yuxuan Li, Xiao Zhai, Yuanbo He, Hongyu Wu, Aimin Hao
https://doi.org/10.2312/pg.20251267 (Pacific Graphics (CCF-B))

Motivation: cutting changes mesh topology on the fly, invalidating the precomputed constraint partitioning and coloring that parallel GPU solvers rely on. Novelty: a parallel constraint-graph re-partitioning and re-coloring scheme that restores solver parallelism instantly, enabling realtime soft-body cutting.

A Unified Particle-Based Solver for non-Newtonian Behaviors Simulation Dec 2023

Chunlei Li, Yang Gao, Jiayi He, Tianwei Cheng, Shuai Li, Aimin Hao

https://doi.org/10.1109/TVCG.2023.3341453 (TVCG (CCF-A, SCI Q1) | https://chunleili.github.io/project-page-nonNewton/)

Motivation: non-Newtonian behaviors in blood, melting chocolate, and water-starch mixtures span fluid-like and solid-like regimes that single-purpose solvers cannot capture consistently. Novelty: a unified SPH solver using a generalized Maxwell model for unified physical modeling of viscosity, elasticity, and plasticity, reproducing shear-thinning/thickening and viscoelastic effects in one framework.

Comparison between Two Eulerian-Lagrangian Methods: CFD-DEM & MPPIC on the biomass gasification in a fluidized bed Feb 2021

Chunlei Li, Qitai Eri

https://doi.org/10.1007/s13399-021-01384-2 (Biomass Conversion and Biorefinery (SCI Q2))

(Work during master's studies)

Comparative Study of Three Modified sCO2 Brayton Recompression Cycles Based on Energy and Exergy Analysis with GA Optimization Jan 2021

Chunlei Li, Qitai Eri

https://doi.org/10.1504/IJEX.2021.115652 (International Journal of Exergy (SCI Q4))

(Work during master's studies)

Multi-objective Optimization of sCO_2, sCO_2/tCO_2 Cycles Based on Energy-Exergy-Economy Balanced Analysis Apr 2022

Chunlei Li, Qitai Eri

https://doi.org/10.1504/IJEX.2022.122308 (International Journal of Exergy (SCI Q4))

(Work during master's studies)

Internships and Projects

R&D, Zeno Tech – Online June 2022 – Dec 2022

- Intern: R&D of the PBD method in the DCC software using C++.

R&D, Taichi Graphics – Beijing Feb 2023 – Sept 2023

- Intern: R&D of the PBD method. Supervised by Dr. Tiantian Liu, development of PBD solver.

Industry Collaboration Project, Alibaba – Beijing May 2025 – Nov 2025

- Developed a CUDA-based Houdini HDK muscle solver node using the MGPBD algorithm, achieving a 4-10x speedup over Vellum at equivalent animation quality.

Intern, miHoYo Sept 2026 – present

- AI physics for games, focusing on neural cloth simulation.

Awards

Top 10 Outstanding Graduate Students Candidate of Beihang University: Beihang University

Outstanding Graduate of Beihang University: Beihang University

Outstanding Graduate of NCEPU (Beijing): NCEPU (Beijing)

Skills

Languages: CET Band 6: 578, CET Band 4: 560, TOEFL: 97, GRE: 323+3.5

DCC Software: Houdini(5 years user)

Physics Engines: Newton (PR contributor), MuJoCo, PhysX